Nature of the complex	Low spin (Spin paired)
Electronic configuration of central metal ion	$t^5_{2g}\, e^0_g$
Magnetic property	No. of unparied electrons = 1; Hence paramagnetic
Magnetic moment (Using spin only formula)	$\mu_s = \sqrt{n(n+2)} = \sqrt{1(1+2)} = 1.732$ BM

Colour of the complex and crystal field splitting energy:

Most of the transition metal complexes are coloured. A substance exhibits colour when it absorbs the light of a particular wavelength in the visible region and transmit the rest of the visible light. When this transmitted light enters our eye, our brain recognises its colour. The colour of the transmitted light is given by the complementary colour of the absorbed light. For example, the hydrated copper(II) ion is blue in colour as it absorbs orange light, and transmit its complementary colour, blue. A list of absorbed wavelength and their complementary colour is given in the following table.

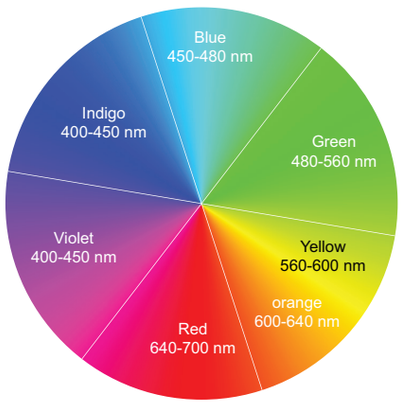


Figure 5.15 Colour Wheel - Complementary colours are shown on opposite sides.

Wave length(λ) of absorbed light (Å)	Wave number(ν) of the absorbed light (cm-1)	Colour of absorbed light	Observed Colour
4000	25000	Violet	Yellow
4750	21053	Blue	Orange
5100	19608	Green	Red
5700	17544	Yellow	Violet
5900	16949	Orange	Blue
6500	15385	Red	Green

The observed colour of a coordination compound can be explained using crystal field theory. We learnt that the ligand field causes the splitting of d orbitals of the central metal atom into two sets (t_{2g} and e_g). When the white light falls on the complex ion, the central metal ion absorbs visible light corresponding to the crystal filed splitting energy and transmits rest of the light which is responsible for the colour of the complex.

XII U5 Coordination jagan.indd 159 04-01-2021 13:24:17